

A maximal abelian subalgebra containing a self-adjoint element

MathlibAnnex.Analysis.CStarAlgebra.exists_maximalAbelian_containing_isSelfAdjoint

theorem

Starts with the algebra generated by the given element and extends it by inclusion-maximality.

Statement

Let A be a unital complex C^* -algebra and let $x \in A$ satisfy $x^* = x$. There is a unital $*$ -subalgebra $D \subseteq A$ which contains x and is maximal among commutative unital $*$ -subalgebras.

Assumptions

There is no assumption that $x \neq 0$, that A is nonzero or separable, or that a representation has been chosen.

Conclusion

The same D is commutative, inclusion-maximal among commutative unital $*$ -subalgebras, and contains the specified x .

Proof route

Self-adjointness supplies normality; apply the normal-element containment theorem, whose maximality comes from Zorn's lemma.

▼ Proof steps

1. Self-adjointness gives $xx^* = x^*x$, so x is normal. The normal-element theorem Maximal abelian containment for a normal element therefore applies to this x . Its starting subalgebra is the closed unital $*$ -subalgebra $E = C^*(1, x)$; normality makes E commutative and $x \in E$.
2. Consider all commutative unital $*$ -subalgebras containing E , ordered by inclusion. This set is nonempty because it contains E . A nonempty chain has the union as its algebraic star-subalgebra upper bound: finitely many elements needed in an algebra operation lie together in one chain member, and any two elements commute there. This is the directed supremum used in the source. An empty chain is bounded by E . Zorn's lemma gives a maximal such D .
3. If a commutative unital $*$ -subalgebra F contains D , then it also contains E , so the maximality just obtained forces $F \subseteq D$. Thus D is maximal abelian in the full sense, and $x \in E \subseteq D$. The separate result Maximal abelianness implies norm closedness also gives norm closedness: $\overline{D}$ is a commutative $*$ -subalgebra containing D , so maximality forces $\overline{D} = D$.

Main citations

- Exact theorem and proof
- Maximal abelian containment for a normal element
- Maximal abelianness implies norm closedness
- The maximal abelian predicate

Lean source signature (exact)

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theorem exists_maximalAbelian_containing_isSelfAdjoint
  (x : A) (hx : IsSelfAdjoint x) :
  ∃ D : StarSubalgebra ℂ A, IsMaximalAbelian D ∧ x ∈ D
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In the source	Mathematical meaning
$(x : A)$	The specified element of the unital complex C^* -algebra A .
$(hx : \text{IsSelfAdjoint } x)$	The equation $x^* = x$.
$\exists D : \text{StarSubalgebra } \mathbb{C} A$	One unital complex $*$ -subalgebra $D \subseteq A$ is chosen.
$\text{IsMaximalAbelian } D$	That D is commutative, and every commutative unital $*$ -subalgebra containing it equals it.
$\wedge x \in D$	The same D contains the original specified element x .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.exists_maximalAbelian_containing_isSelfAdjoint

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